London Mathematical Society
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Analytic Semigroups and Semilinear Initial Boundary Value Problems

Kazuaki Taira

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London Mathematical Society Lecture Note Series. 223

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CAMBRIDGE
UNIVERSITY PRESS

Published by the Press Syndicate of the University of Cambridge
The Pitt Building, Trumpington Street, Cambridge CB2 1RP
40 West 20th Street, New York, NY 10011-4211, USA
10 Stamford Road, Oakleigh, Melbourne 3166, Australia

First published 1995

Library of Congress cataloging in publication data available

British Library cataloguing in publication data available

ISBN 0 521 55603 1 paperback

Transferred to digital printing 2004

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TO MY MOTHER

PREFACE

This monograph is devoted to the functional analytic approach to initial boundary value problems for semilinear parabolic differential equations. First we study non-homogeneous boundary value problems for second-order elliptic differential operators, in the framework of Sobolev spaces of L^p style, which include as particular cases the Dirichlet and Neumann problems. We prove that these boundary value problems provide an example of analytic semigroups in the L^p topology. The essential point in the proof is to define a function space which is a tool well suited to investigating our boundary conditions. By virtue of the theory of analytic semigroups, one can apply this result to the study of the initial boundary value problems for semilinear parabolic differential equations in the framework of L^p spaces.

This monograph grew out of a set of lecture notes "On initial boundary value problems for semilinear parabolic differential equations" for graduate courses given at the University of Tsukuba in winter 1994/95. In order to make this monograph accessible to a broad readership, I have tried to start from scratch. In the preparatory chapters, we even prove fundamental results like a generation theorem for analytic semigroups in functional analysis and Sobolev imbeddings theorems in partial differential equations. Furthermore, we summarize the basic definitions and results about the L^p theory of pseudo-differential operators which is considered as a modern theory of potentials. The L^p theory of pseudo-differential operators forms a most convenient tool in the study of elliptic boundary value problems in the framework of Sobolev spaces of L^p style. The material in these preparatory chapters is given for completeness, to minimize the necessity of consulting too many outside references. This makes the monograph fairly self-contained.

This work was begun at the University of Turin and the University of Bologna in May 1988 under the sponsorship of the Italian "Consiglio Nazionale delle Ricerche" and a major part of the work was done at the University of the Philippines in the course of the JSPS-DOST exchange program from January 1989 to March 1989 while I was on leave from the University of Tsukuba. I take this opportunity to express my gratitude to all these institutions for their hospitality and support. Thanks are also due to the editorial staff of Cambridge University Press for their unfailing helpfulness and cooperation during the production of the book.

I hope that this monograph will lead to a better insight into the study

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of initial boundary value problems for semilinear parabolic differential equations. For probabilistic information on the topics discussed here, I would like to call attention to my previous book *Boundary Value Problems and Markov Processes*, Lecture Notes in Mathematics, No. 1499, Springer-Verlag, 1991.

Higashi-Hiroshima, June 1995

Kazuaki Taira

INTRODUCTION AND RESULTS

Let Ω be a bounded domain of Euclidean space $\mathbf{R}^n$, with C^∞ boundary Γ; its closure $\overline{\Omega} = \Omega \cup \Gamma$ is an n-dimensional, compact C^∞ manifold with boundary. We let

$$A = \sum_{i=1}^{n} \frac{\partial}{\partial x_i}\left(\sum_{j=1}^{n} a^{ij}(x)\frac{\partial}{\partial x_j}\right) + \sum_{i=1}^{n} b^i(x)\frac{\partial}{\partial x_i} + c(x)$$

be a second-order *elliptic* differential operator with real C^∞ coefficients on $\overline{\Omega}$ such that:

(1) $a^{ij}(x) = a^{ji}(x)$, $x \in \Omega$, $1 \leq i, j \leq n$.

(2) There exists a constant $a_0 > 0$ such that

$$\sum_{i,j=1}^{n} a^{ij}(x)\xi_i\xi_j \geq a_0|\xi|^2, \quad x \in \overline{\Omega},\ \xi \in \mathbf{R}^n.$$

(3) $c(x) \leq 0$ in Ω, and c does not vanish *identically* in Ω.

We consider the following boundary value problem: Given functions f and φ defined in Ω and on Γ, respectively, find a function u in Ω such that

$$(*) \qquad \begin{cases} Au = f & \text{in } \Omega, \\ Bu := a\frac{\partial u}{\partial \boldsymbol{\nu}} + bu\big|_\Gamma = \varphi & \text{on } \Gamma. \end{cases}$$

Here:

(1) a and b are real-valued, C^∞ functions on Γ.

(2) $\partial/\partial\boldsymbol{\nu}$ is the conormal derivative associated with the operator A:

$$\frac{\partial}{\partial \boldsymbol{\nu}} = \sum_{i,j=1}^{N} a^{ij} n_j \frac{\partial}{\partial x_i},$$

$\mathbf{n} = (n_1, n_2, \cdots, n_n)$ being the unit exterior normal to the boundary Γ (see

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Figure 1).

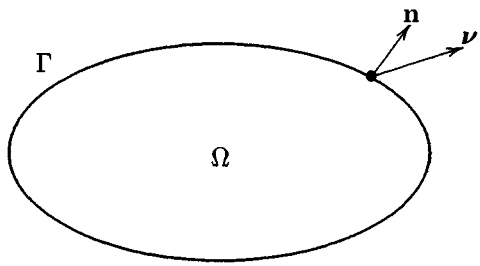

Figure 1

We remark that if $a \equiv 1$ and $b \equiv 0$ on Γ (resp. $a \equiv 0$ and $b \equiv 1$ on Γ), then the boundary condition B is the so-called Neumann (resp. Dirichlet) condition. It is easy to see that problem $(*)$ is non-degenerate (or coercive) if and only if either $a \neq 0$ on Γ or $a \equiv 0$ and $b \neq 0$ on Γ.

The first purpose of this book is to prove an existence and uniqueness theorem for problem $(*)$ under the condition $a \geq 0$ on Γ in the framework of Sobolev spaces of L^p style. The essential point is to define a function space which is a suitable tool to investigate the degenerate boundary condition B.

If $1 \leq p < \infty$, we let

$$L^p(\Omega) = \text{the space of (equivalence classes of) Lebesgue measurable functions } u \text{ on } \Omega \text{ such that } |u|^p \text{ is integrable on } \Omega.$$

The space $L^p(\Omega)$ is a Banach space with the norm

$$\|u\|_p = \left(\int_\Omega |u(x)|^p dx\right)^{1/p}.$$

If s is a positive integer, we define the Sobolev space

$$H^{s,p}(\Omega) = \text{the space of (equivalence classes of) functions } u \in L^p(\Omega) \text{ whose derivatives } D^\alpha u,\ |\alpha| \leq s, \text{ in the sense of distributions are in } L^p(\Omega).$$

The space $H^{s,p}(\Omega)$ is a Banach space with the norm

$$\|u\|_{s,p} = \left(\sum_{|\alpha|\leq s} \int_\Omega |D^\alpha u(x)|^p dx\right)^{1/p}.$$

Furthermore we let

$$B^{s-1/p,p}(\Gamma) = \text{the space of the boundary values } \varphi \text{ of functions } u \in H^{s,p}(\Omega).$$

In the space $B^{s-1/p,p}(\Gamma)$, we introduce a norm

$$|\varphi|_{s-1/p,p} = \inf \|u\|_{s,p},$$

where the infimum is taken over all functions $u \in H^{s,p}(\Omega)$ which equal φ on the boundary. The space $B^{s-1/p,p}(\Gamma)$ is a Banach space with respect to this norm $|\cdot|_{s-1/p,p}$; more precisely, it is a Besov space (cf. [BL], [Tr]).

We introduce a subspace of $B^{s-1-1/p,p}(\Gamma)$ which is associated with the boundary condition

$$Bu = a\frac{\partial u}{\partial \boldsymbol{\nu}} + bu\bigg|_{\Gamma} = 0 \quad \text{on } \Gamma.$$

We let

$$B_*^{s-1-1/p,p}(\Gamma) = \left\{\varphi = a\varphi_1 + b\varphi_2 : \varphi_1 \in B^{s-1-1/p,p}(\Gamma),\ \varphi_2 \in B^{s-1/p,p}(\Gamma)\right\},$$

and define

$$|\varphi|^*_{s-1-1/p,p} = \inf\left\{|\varphi_1|_{s-1-1/p,p} + |\varphi_2|_{s-1/p,p} : \varphi = a\varphi_1 + b\varphi_2\right\}.$$

Then it is easy to verify that the space $B_*^{s-1-1/p,p}(\Gamma)$ is a Banach space with respect to the norm $|\cdot|^*_{s-1-1/p,p}$. We remark that the space $B_*^{s-1-1/p,p}(\Gamma)$ is an "interpolation space" between the spaces $B^{s-1/p,p}(\Gamma)$ and $B^{s-1-1/p,p}(\Gamma)$. More precisely, we have

$$B_*^{s-1-1/p,p}(\Gamma) = B^{s-1/p,p}(\Gamma) \quad \text{if } a \equiv 0 \text{ on } \Gamma,$$
$$B_*^{s-1-1/p,p}(\Gamma) = B^{s-1-1/p,p}(\Gamma) \quad \text{if } a > 0 \text{ on } \Gamma,$$

and, for general a, the continuous injections

$$B^{s-1/p,p}(\Gamma) \subset B_*^{s-1-1/p,p}(\Gamma) \subset B^{s-1-1/p,p}(\Gamma).$$

Now we can state our existence and uniqueness theorem for problem $(*)$ (cf. [Um, Theorem 1]):

Theorem 1. *Let $1 < p < \infty$ and $s > 1+1/p$. Assume that the following two conditions (H.1) and (H.2) are satisfied:*

(H.1) $a(x') \geq 0$ and $b(x') \geq 0$ on Γ.

(H.2) $b(x') > 0$ on $\Gamma_0 = \{x' \in \Gamma : a(x') = 0\}$.

Then the mapping

$$(A, B) : H^{s,p}(\Omega) \longrightarrow H^{s-2,p}(\Omega) \oplus B_*^{s-1-1/p,p}(\Gamma)$$

is an algebraic and topological isomorphism. *In particular, for any $f \in H^{s-2,p}(\Omega)$ and any $\varphi \in B_*^{s-1-1/p,p}(\Gamma)$, there exists a unique solution $u \in H^{s,p}(\Omega)$ of problem $(*)$.*

The second purpose of this book is to study problem $(*)$ from the point of view of analytic semigroup theory in functional analysis. The generation theorem for analytic semigroups is well established in the non-degenerate case in the L^p topology (cf. [Fr]). We shall generalize this generation theorem for analytic semigroups to the *degenerate* case.

First we state a generation theorem for analytic semigroups in the L^p topology. We associate with problem $(*)$ an unbounded linear operator $\mathfrak{A}$ from $L^p(\Omega)$ into itself as follows:

(a) The domain of definition $D(\mathfrak{A})$ of $\mathfrak{A}$ is the set

$$D(\mathfrak{A}) = \left\{u \in H^{2,p}(\Omega) : Bu = 0\right\}.$$

(b) $\mathfrak{A}u = Au$, $u \in D(\mathfrak{A})$.

The next theorem is an L^p version of [Ta1, Theorem 1]:

Theorem 2. *Let $1 < p < \infty$. If conditions (H.1) and (H.2) are satisfied, then we have the following:*

(i) For every $0 < \varepsilon < \pi/2$, there exists a constant $r(\varepsilon) > 0$ such that the resolvent set of $\mathfrak{A}$ contains the set

$$\Sigma(\varepsilon) = \left\{\lambda = r^2 e^{i\theta} : r \geq r(\varepsilon), -\pi + \varepsilon \leq \theta \leq \pi - \varepsilon\right\},$$

and that the resolvent $(\mathfrak{A} - \lambda I)^{-1}$ satisfies the estimate

$$\left\|(\mathfrak{A} - \lambda I)^{-1}\right\| \leq \frac{c(\varepsilon)}{|\lambda|}, \quad \lambda \in \Sigma(\varepsilon),$$

where $c(\varepsilon) > 0$ is a constant depending on ε.

(ii) The operator $\mathfrak{A}$ generates a semigroup $U(z)$ on the space $L^p(\Omega)$ which is analytic *in the sector $\Delta_\varepsilon = \{z = t + is : z \neq 0, |\arg z| < \pi/2 - \varepsilon\}$ for any $0 < \varepsilon < \pi/2$ (see Figure 2).*

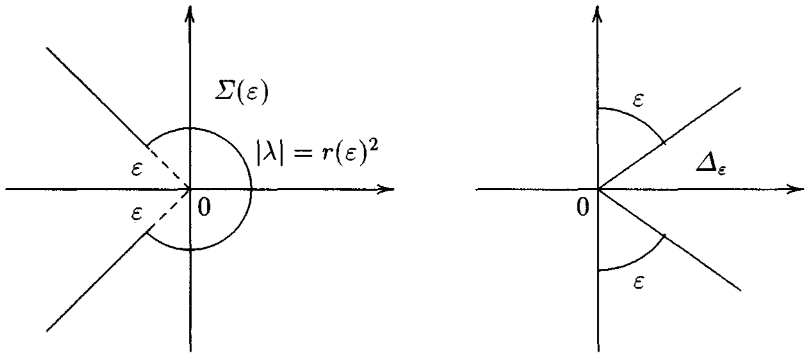

Figure 2

Next, as an application of Theorem 2, we consider the following *semilinear* initial boundary value problem: Given functions f and u_0 defined in $\Omega \times [0,T) \times \mathbf{R} \times \mathbf{R}^N$ and in Ω, respectively, find a function u in $\Omega \times [0,T)$ such that

(**)

$$\begin{cases} (\frac{\partial}{\partial t} - A)u(x,t) = f(x,t,u,\operatorname{grad} u) & \text{in } \Omega \times (0,T), \\ Bu(x',t) := a(x')\frac{\partial u}{\partial \nu}(x',t) + b(x')u(x',t)\big|_{\Gamma\times[0,T)} = 0 & \text{on } \Gamma \times [0,T), \\ u(x,0) = u_0(x) & \text{in } \Omega. \end{cases}$$

By using the operator $\mathfrak{A}$, one can formulate problem (**) in terms of the *abstract Cauchy problem* in the space $L^p(\Omega)$ as follows:

$$(**') \qquad \begin{cases} \frac{du}{dt} = \mathfrak{A}u(t) + F(t,u(t)), \ 0 < t < T, \\ u|_{t=0} = u_0. \end{cases}$$

Here $u(t) = u(\cdot,t)$ and $F(t,u(t)) = f(\cdot,t,u(t),\operatorname{grad} u(t))$ are functions defined on the interval $[0,T)$, taking values in the space $L^p(\Omega)$.

First we consider the case $p > n$:

Theorem 3. *Assume that conditions (H.1) and (H.2) are satisfied. Let $n < p < \infty$ and let $f(x,t,u,\xi)$ be a locally Lipschitz continuous function of all its variables with the possible exception of the x variable. Then, for every function u_0 of $D(\mathfrak{A})$, problem (**$'$) has a unique local solution $u \in C([0,T']; L^p(\Omega)) \cap C^1((0,T'); L^p(\Omega))$ where $T' = T'(p,u_0) > 0$.*

Here $C([0,T']; L^p(\Omega))$ denotes the space of continuous functions on $[0,T']$ taking values in $L^p(\Omega)$, and $C^1((0,T'); L^p(\Omega))$ denotes the space of continuously differentiable functions on $(0,T')$ taking values in $L^p(\Omega)$, respectively.

In the case $p < n$, the domain $D(\mathfrak{A})$ is large compared with the case $n < p < \infty$. Hence we must impose some growth conditions on the function f:

Theorem 4. *Assume that conditions (H.1) and (H.2) are satisfied. Let* $n/2 < p < n$ *and let* $f(x, t, u, \xi)$ *be a locally Lipschitz continuous function of all its variables with the possible exception of the* x *variable. Further assume that there exist a non-negative continuous function* $\rho(t, r)$ *on* $\mathbf{R} \times \mathbf{R}$ *and a constant* $1 \leq \gamma < n/(n-p)$ *such that:*

(a) $|f(x, t, u, \xi)| \leq \rho(t, |u|)(1 + |\xi|^\gamma)$.
(b) $|f(x, t, u, \xi) - f(x, s, u, \xi)| \leq \rho(t, |u|)\,(1 + |\xi|^\gamma)\,|t - s|$.
(c) $|f(x, t, u, \xi) - f(x, t, u, \eta)| \leq \rho(t, |u|)\left(1 + |\xi|^{\gamma-1} + |\eta|^{\gamma-1}\right)|\xi - \eta|$.
(d) $|f(x, t, u, \xi) - f(x, t, v, \xi)| \leq \rho(t, |u| + |v|)\,(1 + |\xi|^\gamma)\,|u - v|$.

Then, for every function u_0 *of* $D(\mathfrak{A})$, *problem* $(**')$ *has a unique local solution* $u \in C\left([0, T']; L^p(\Omega)\right) \cap C^1\left((0, T'); L^p(\Omega)\right)$ *where* $T' = T'(p, u_0) > 0$.

Theorems 3 and 4 are a generalization of Pazy [Pa, Section 8.4, Theorems 4.4 and 4.5] to the degenerate case.

The rest of this book is organized as follows.

Chapter 1 is devoted to a review of standard topics from the theory of analytic semigroups which forms a functional analytic background for the proof of Theorems 2, 3 and 4. First we prove a generation theorem of analytic semigroups (Theorem 1.2). Next we consider the abstract linear and semilinear Cauchy problems, and give a complete proof of local existence and uniqueness theorems (Theorems 1.16 and 1.18).

In Chapter 2, we study the imbedding characteristics of Sobolev spaces of L^p style that render these spaces so useful in the study of partial differential equations. We give a complete proof of the most important of the imbedding properties of the spaces $H^{m,p}(\mathbf{R}^n)$ (Theorems 2.15 and 2.18 and Corollary 2.22).

In Chapter 3, we present a brief description of the basic concepts and results of the L^p theory of pseudo-differential operators which may be considered as a generalization of the classical potential theory. In particular, we state a Besov-space boundedness theorem (Theorem 3.17) due to Bourdaud [Bo] and a criterion for hypoellipticity for pseudo-differential operators (Theorem 3.19) due to Hörmander [Ho].

In Chapter 4, we study the boundary value problem $(*)$ in the framework of Sobolev spaces of L^p style, by using the L^p theory of pseudo-differential operators. We prove that problem $(*)$ can be reduced to the study of a pseudo-differential operator T on the boundary Γ (Theorems 4.10-4.12).

In Chapter 5, we study the pseudo-differential operator T in question, and prove Theorem 1. In Section 5.1 we prove a regularity theorem for problem $(*)$ (Theorem 5.1). More precisely, we prove that if conditions (H.1) and (H.2) are satisfied, one can construct a parametrix S for T in the Hörmander class $L^0_{1,1/2}(\Gamma)$ (Lemma 5.2), and then apply a Besov-space boundedness theorem to the parametrix S to obtain the regularity theorem for problem $(*)$. Section 5.2 is devoted to a uniqueness theorem for problem $(*)$ (Theorem 5.5). In the proof we make good use of the maximum principle stated in the